

Redis

- Redis는 데이터를 주로 메모리에 저장하는 키-값 기반 데이터 저장소입니다.
- 디스크보다 빠른 메모리에서 데이터를 읽고 쓰기 때문에 빠른 응답이 필요한 캐시, 세션, 카운터, 랭킹, 메시지 처리 등에 자주 사용합니다.
- Redis의 이름은 **Remote Dictionary Server**에서 왔습니다. 단순한 문자열뿐 아니라 List, Set, Sorted Set, Hash 같은 여러 자료구조를 제공합니다.

관계형 데이터베이스와 비교

구분	관계형 데이터베이스	Redis
데이터 위치	주로 디스크	주로 메모리
데이터 표현	테이블, 행, 열	키와 자료구조
조회 방식	SQL	Redis 명령어
강점	복잡한 조회, 관계, 영구 보관	빠른 조회, 만료, 실시간 처리
대표 용도	주문·결제 원본 데이터	캐시·세션·랭킹·카운터

Redis가 빠르다고 해서 관계형 데이터베이스를 모두 대체하는 것은 아닙니다. 주문과 결제 같은 중요한 원본 데이터는 관계형 데이터베이스에 저장하고, 자주 조회하는 결과를 Redis에 캐시하는 방식으로 함께 사용하는 경우가 많습니다.

Redis의 주요 특징

1. 메모리 기반이라 읽기와 쓰기가 빠릅니다.
2. 키마다 만료 시간을 설정할 수 있습니다.
3. 다양한 자료구조와 원자적 명령을 제공합니다.
4. RDB 또는 AOF 방식으로 데이터를 디스크에 남길 수 있습니다.
5. 복제와 클러스터를 통해 가용성과 확장성을 높일 수 있습니다.

메모리는 유한하며 프로세스 종료이나 장애도 발생할 수 있습니다.

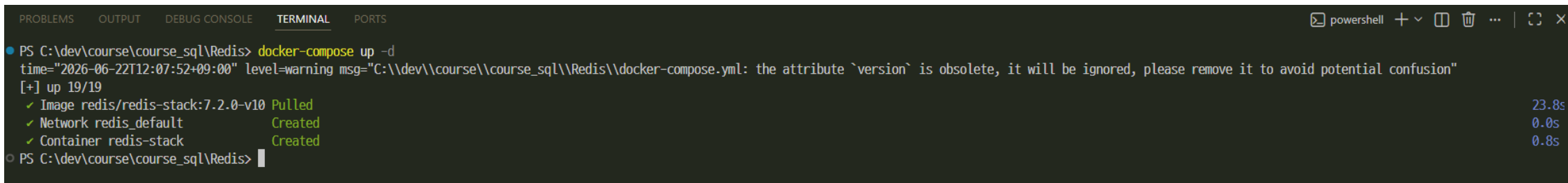
Redis를 사용할 때는 데이터의 중요도, 메모리 한도, 영속성 정책을 함께 결정해야 합니다.

Redis 설치 및 접속

Redis 설치 with Docker

Redis 폴더 이동 후 명령어 실행

```
docker-compose up -d
```



```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\dev\course\course_sql\Redis> docker-compose up -d
time="2026-06-22T12:07:52+09:00" level=warning msg="C:\dev\course\course_sql\Redis\docker-compose.yml: the attribute `version` is obsolete, it will be ignored, please remove it to avoid potential confusion"
[+] up 19/19
  ✓ Image redis/redis-stack:7.2.0-v10 Pulled 23.8s
  ✓ Network redis_default Created 0.0s
  ✓ Container redis-stack Created 0.8s
PS C:\dev\course\course_sql\Redis> 
```

 **docker**.desktop

PERSONAL



G





Containers

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Container CPU usage 

1.56% / 1600% (16 CPUs available)

Container memory usage 

294MB / 14.87GB

[Show charts](#)

 Search

 Only running

	Name	Container ID	Image	Port(s)	Actions
 > 	postgres	-	-	-	  
 v 	redis	-	-	-	  
	stack	fb3ccc190b6c	redis/redis	6379:6379  Show all ports (2)	  

Redis UI 접속

<http://localhost:8001/>

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Select to activate all listed options.

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☐ **Enable Analytics**

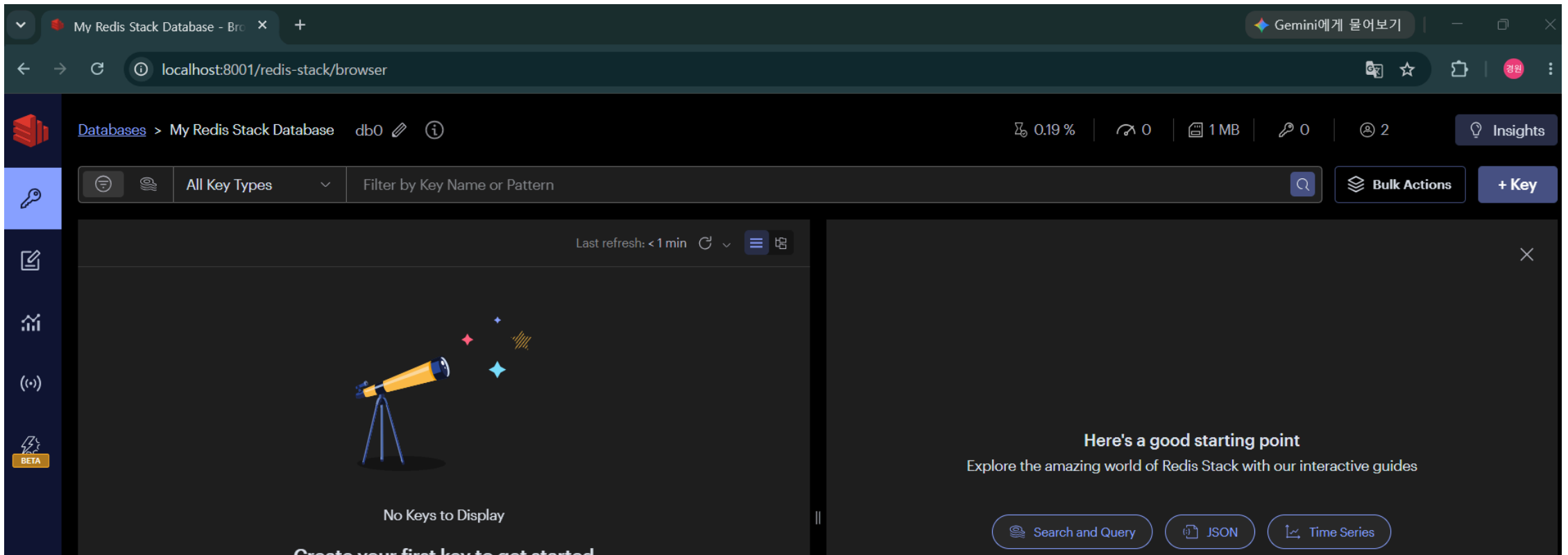
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Redis Server 접속

 **docker desktop** PERSONAL

      G 

Containers [Give feedback](#)

Container CPU usage ⓘ
1.05% / 1600% (16 CPUs available)

Container memory usage ⓘ
298.4MB / 14.87GB

[Show charts](#)

  Only running

☐	Name	Container ID	Image	Port(s)	Actions
☐ > ●	postgres	-	-	-	  
☐ v ●	redis	-	-	-	  
☐ ●	stack ←	fb3ccc190b6c	redis/redis	6379:6379  Show all ports (2)	  

redis-cli 를 실행한 뒤 다음 명령을 입력합니다.

The screenshot shows the Docker Desktop interface. The top bar is blue with the 'docker desktop' logo and 'PERSONAL' label. The left sidebar contains icons for containers, volumes, networks, and other resources. The main area displays the 'redis-stack' container details, including its ID 'fb3ccc190b6c' and image 'redis/redis-stack:7.2.0-v10'. The 'STATUS' is 'Running (5 minutes ago)'. Below the container details, there are tabs for 'Logs', 'Inspect', 'Bind mounts', 'Exec', 'Files', and 'Stats'. The 'Exec' tab is selected and highlighted with a red box. A terminal window is open in the 'Exec' tab, showing the prompt '# redis-cli' and the IP address '127.0.0.1:6379>'. A red arrow points to the terminal input area. On the right side of the terminal, there are search and copy icons. Above the terminal, there are buttons for 'Stop', 'Start', 'Restart', and 'Delete'.

Containers / redis-stack

redis-stack

< fb3ccc190b6c [redis/redis-stack:7.2.0-v10](#) **STATUS** Running (5 minutes ago)

[6379:6379](#) [8001:8001](#)

Logs Inspect Bind mounts **Exec** Files Stats ☐ Debug mode [Open in external terminal](#)

```
# redis-cli  
127.0.0.1:6379> 
```

Redis 명령어

```
# Redis 서버가 정상적으로 동작하는지 확인 (응답: PONG)
PING
```

```
# greeting이라는 Key에 "Hello Redis" 문자열 저장
SET greeting "Hello Redis"
```

```
# greeting Key의 값을 조회
GET greeting
```

```
# greeting Key 삭제
DEL greeting
```

```
# 삭제된 greeting Key 조회
GET greeting
```

redis-stack



fb3ccc190b6c [redis/redis-stack](#)
[6379:6379](#) [8001:8001](#)

Logs

Inspect

Bind mounts

Exec

```
127.0.0.1:6379> PING
PONG
127.0.0.1:6379> SET greeting "Hello Redis"
OK
127.0.0.1:6379> GET greeting
"Hello Redis"
127.0.0.1:6379> DEL greeting
(integer) 1
127.0.0.1:6379> GET greeting
(nil)
127.0.0.1:6379> 
```

명령 구조 읽기

명령어 키 값 또는 옵션

```
SET      greeting      "Hello Redis"
```

- SET : 수행할 작업
- greeting : 데이터를 찾기 위한 키
- "Hello Redis" : 저장할 값

서버 정보 확인

DBSIZE : 현재 선택된 Redis 데이터베이스(DB)에 저장된 Key의 개수를 반환하는 명령어
Redis 버전, 실행 모드, 사용 메모리 정도만 찾아봅니다.

The screenshot shows the Docker Desktop interface. The top bar is blue with the 'docker.desktop' logo and a 'PERSONAL' tab. On the right, there are icons for search, help, notifications (4), Docker Hub, settings, a grid of containers, and a user profile 'G'. The left sidebar contains icons for containers, volumes, networks, services, and other Docker resources. The main area displays the details for a container named 'redis-stack'. It shows the container ID 'fb3ccc190b6c', the image 'redis/redis-stack:7.2.0-v10', and ports '6379:6379' and '8001:8001'. The status is 'Running (5 minutes ago)'. Below this, there are tabs for 'Logs', 'Inspect', 'Bind mounts', 'Exec' (which is active), 'Files', and 'Stats'. The 'Exec' tab shows a terminal window with the command 'DBSIZE' and its output '(integer) 0'. There are also buttons for 'Stop', 'Start', 'Restart', and 'Remove'.

Containers / redis-stack

redis-stack

fb3ccc190b6c redis/redis-stack:7.2.0-v10

6379:6379 8001:8001

STATUS
Running (5 minutes ago)

Logs Inspect Bind mounts **Exec** Files Stats

127.0.0.1:6379> DBSIZE
(integer) 0
127.0.0.1:6379>

Debug mode Open in external terminal

INFO server : Redis 서버 자체의 상태 정보(Server Information)만 조회하는 명령어

The screenshot shows the Docker Desktop interface. The top bar is blue with the 'docker desktop' logo and 'PERSONAL' label. The left sidebar contains icons for containers, volumes, networks, and other resources. The main area displays the details for a container named 'redis-stack'. The container is running, as indicated by the 'STATUS' section which says 'Running (5 minutes ago)'. The container ID is 'fb3ccc190b6c' and the image is 'redis/redis-stack:7.2.0-v10'. The ports are '6379:6379' and '8001:8001'. The 'Exec' tab is selected, showing the command '127.0.0.1:6379> INFO server' and its output:

```
127.0.0.1:6379> INFO server
# Server
redis_version:7.2.4
redis_git_sha1:00000000
redis_git_dirty:0
redis_build_id:3a23a5ac67fe7008
redis_mode:standalone
os:Linux 6.6.87.2-microsoft-standard-WSL2 x86_64
```

INFO memory : Redis의 메모리 사용 현황을 조회하는 명령어

The screenshot shows the Docker Desktop interface. The top bar is blue with the Docker logo, 'docker.desktop PERSONAL', and various icons including a search icon, a help icon, a notification bell with a red '4', a Docker icon, a settings gear, a grid icon, and a user profile 'G'. The left sidebar contains icons for containers, volumes, networks, and other Docker resources. The main area displays the 'redis-stack' container details. The container name is 'redis-stack', its ID is 'fb3ccc190b6c', and it is using the image 'redis/redis-stack:7.2.0-v10'. The status is 'Running (5 minutes ago)'. Below the container details, there are tabs for 'Logs', 'Inspect', 'Bind mounts', 'Exec' (which is selected), 'Files', and 'Stats'. The 'Exec' tab shows the command '127.0.0.1:6379> INFO memory' and its output. The output shows memory usage statistics: used_memory:1515840, used_memory_human:1.45M, used_memory_rss:23068672, used_memory_rss_human:22.00M, used_memory_peak:1615296, and used_memory_peak_human:1.54M. On the right side of the terminal output, there are icons for search, copy, and delete.

Containers / redis-stack

redis-stack

<  fb3ccc190b6c  redis/redis-stack:7.2.0-v10

6379:6379  8001:8001 

STATUS
Running (5 minutes ago)

Logs Inspect Bind mounts **Exec** Files Stats

127.0.0.1:6379> INFO memory
Memory
used_memory:1515840
used_memory_human:1.45M
used_memory_rss:23068672
used_memory_rss_human:22.00M
used_memory_peak:1615296
used_memory_peak_human:1.54M

Debug mode [Open in external terminal](#) 